STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-REPRESENTATIONS-HULL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 635-648 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 650-687, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_examples-defos.json (pb=1589, stmt 635-648, proof 650-687).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D3J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 483–494. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06J8: sources/stacks-examples-defos/pinned-original-source.tex, source lines 687–687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06J7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 687–687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06IZ: sources/stacks-examples-defos/pinned-original-source.tex, source lines 687–687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06JM: sources/stacks-examples-defos/pinned-original-source.tex, source lines 687–687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVS: sources/stacks-examples-defos/pinned-original-source.tex, source lines 526–532. Cited by the retained statement or proof

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but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVT: sources/stacks-examples-defos/pinned-original-source.tex, source lines 564–578. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0D3J assume Γ finitely generated. Let $\rho_0 : \Gamma \to GL_k(V)$ be a finite dimensional representation. Assume Λ is a complete local ring with residue field k (the classical case). Then the functor*

$$F : \mathcal{C}_\Lambda \longrightarrow Sets, \quad A \longmapsto \mathrm{Ob}(\mathcal{D}ef_{V,\rho_0}(A))/\cong$$

of isomorphism classes of objects has a hull. If $H^0(\Gamma, End_k(V)) = k$, then F is prorepresentable.

Proof. The existence of a hull follows from Lemmas 0DVS and 0DVT and Formal Deformation Theory, Lemma 06J7 and Remark 06IZ.

Assume $H^0(\Gamma, \mathrm{End}_k(V)) = k$. To see that F is prorepresentable it suffices to show that F is a deformation functor, see Formal Deformation Theory, Theorem 06JM. In other words, we have to show F satisfies (RS). For this we can use the criterion of Formal Deformation Theory, Lemma 06J8. The required surjectivity of automorphism groups will follow if we show that

$$A \cdot \mathrm{id}_M = \mathrm{End}_{A[\Gamma]}(M)$$

for any object (A, M, ρ) of $\mathcal{F}$ such that $M \otimes_A k$ is isomorphic to V as a representation of Γ. Since the left hand side is contained in the right hand side, it suffices to show $\mathrm{length}_A \mathrm{End}_{A[\Gamma]}(M) \leq \mathrm{length}_A A$. Choose pairwise distinct ideals $(0) = I_n \subset \ldots \subset I_1 \subset A$ with $n = \mathrm{length}(A)$. By correspondingly filtering M, we see that it suffices to prove $\mathrm{Hom}_{A[\Gamma]}(M, I_t M/I_{t+1}M)$ has length 1. Since $I_t M/I_{t+1}M \cong M \otimes_A k$ and since any $A[\Gamma]$-module map $M \to M \otimes_A k$ factors uniquely through the quotient map $M \to M \otimes_A k$ to give an element of

$$\mathrm{End}_{A[\Gamma]}(M \otimes_A k) = \mathrm{End}_{k[\Gamma]}(V) = k$$

we conclude. □